

Memorandum

Explicit Engineering Notation for Variational Free Energy and Active Inference Planning

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Purpose

This memorandum summarizes two pieces of notation that make Friston’s Active Inference framework more transparent from the standpoint of nonlinear stochastic state-space modeling, system identification, and finite-horizon stochastic control.

The first part reviews ordinary variational free energy, denoted here by F , in a form that makes clear where modeling assumptions enter. The second part rewrites expected free energy, denoted here by $G(\pi)$, so that the dependence on future stacked state vectors, future observation vectors, and future control inputs is explicit.

1 Nonlinear Stochastic State-Space Model

Let the hidden state be

$$x_k \in \mathbb{R}^n,$$

and let the observation be

$$y_k \in \mathbb{R}^m.$$

A nonlinear stochastic state-space model may be written as

$$x_{k+1} = f(x_k, u_k, w_k), \tag{1}$$

$$y_k = h(x_k, v_k), \tag{2}$$

where u_k is an input or action, w_k is process noise, and v_k is observation noise. Equivalently, one may specify transition and observation densities,

$$p(x_{k+1} \mid x_k, u_k), \tag{3}$$

$$p(y_k \mid x_k). \tag{4}$$

The observation density $p(y_k \mid x_k)$ is where concrete modeling assumptions enter. For example, a Gaussian observation model might be

$$y_k = h(x_k) + v_k, \quad v_k \sim \mathcal{N}(0, R),$$

which implies

$$p(y_k \mid x_k) = \mathcal{N}(y_k; h(x_k), R).$$

2 Variational Free Energy F : Inference About Present or Past Hidden States

Suppose that the hidden variable is x and the observation is y . The generative model is

$$p(y, x) = p(y | x)p(x).$$

The true Bayesian posterior is

$$p(x | y) = \frac{p(y | x)p(x)}{p(y)},$$

where

$$p(y) = \int p(y | x)p(x) dx$$

is the marginal likelihood, or model evidence.

In variational inference, the posterior $p(x | y)$ is approximated by a tractable density $q(x)$. The variational free energy is defined as

$$F[q] = \int q(x) \log \frac{q(x)}{p(y, x)} dx. \quad (5)$$

Using $p(y, x) = p(y | x)p(x)$,

$$F[q] = \int q(x) \log q(x) dx - \int q(x) \log p(y | x) dx - \int q(x) \log p(x) dx \quad (6)$$

$$= D_{\text{KL}}(q(x) \| p(x)) - \mathbb{E}_{q(x)} [\log p(y | x)]. \quad (7)$$

This is often the most useful engineering form:

$$\boxed{F[q] = D_{\text{KL}}(q(x) \| p(x)) - \mathbb{E}_q [\log p(y | x)]}$$

The first term penalizes movement away from the prior. The second term is a negative expected log-likelihood, or data-fit term.

An equivalent form follows from Bayes' rule:

$$F[q] = D_{\text{KL}}(q(x) \| p(x | y)) - \log p(y). \quad (8)$$

Thus

$$F[q] \geq -\log p(y),$$

because KL divergence is nonnegative. The minimum occurs when

$$q(x) = p(x | y).$$

Therefore, minimizing F with respect to q performs approximate Bayesian inference.

Key interpretation

Variational free energy F answers the question:

$$\boxed{\text{What hidden state best explains the observations already obtained?}}$$

It is an inference objective, not a planning objective.

3 Expected Free Energy $G(\pi)$: Planning Over Future Trajectories

Expected free energy $G(\pi)$ is different from F . It is used to evaluate possible future policies.

At time t , let the planning horizon be H . Define a policy as a stacked future control sequence

$$U_\pi = \begin{bmatrix} u_t \\ u_{t+1} \\ \vdots \\ u_{t+H-1} \end{bmatrix}.$$

Under this policy, the future hidden states form the stacked vector

$$X_\pi = \begin{bmatrix} x_{t+1} \\ x_{t+2} \\ \vdots \\ x_{t+H} \end{bmatrix},$$

and the corresponding future observations form

$$Y_\pi = \begin{bmatrix} y_{t+1} \\ y_{t+2} \\ \vdots \\ y_{t+H} \end{bmatrix}.$$

The trajectory is constrained by the future state dynamics

$$x_{k+1} = f(x_k, u_k, w_k), \quad k = t, \dots, t + H - 1,$$

and observation model

$$y_k = h(x_k, v_k), \quad k = t + 1, \dots, t + H.$$

Friston's compact notation $G(\pi)$ suppresses this entire trajectory structure. A more explicit notation would be

$$G(\pi) = G(U_\pi; X_\pi, Y_\pi),$$

with the understanding that X_π and Y_π are random future trajectories induced by the candidate control sequence U_π .

4 Predictive Distributions Under a Policy

Given the current belief $q_t(x_t)$, each policy induces future predictive state distributions recursively:

$$q_\pi(x_{k+1}) = \int p(x_{k+1} \mid x_k, u_k) q_\pi(x_k) dx_k. \quad (9)$$

The predicted future observation distribution is

$$q_\pi(y_k) = \int p(y_k \mid x_k) q_\pi(x_k) dx_k. \quad (10)$$

Thus $G(\pi)$ is computed using probability distributions over future states and observations, not only point predictions.

5 Preference Distribution and Setpoints

Active Inference replaces an ordinary reward function with a preference distribution over future observations,

$$p_{\text{pref}}(y_k).$$

A useful engineering interpretation is that $p_{\text{pref}}(y_k)$ defines desired setpoints and tolerances for observed variables.

For example, if

$$p_{\text{pref}}(y_k) = \mathcal{N}(y^*, \Sigma),$$

then

$$-\log p_{\text{pref}}(y_k) = \frac{1}{2}(y_k - y^*)^T \Sigma^{-1}(y_k - y^*) + \text{constant}. \quad (11)$$

Thus the preference term becomes a quadratic tracking cost around the desired setpoint y^* .

6 Explicit Trajectory Form of Expected Free Energy

One useful engineering decomposition of expected free energy is

$$G(\pi) = G_{\text{pref}}(\pi) + G_{\text{info}}(\pi).$$

6.1 Preference or Extrinsic Component

The preference cost accumulated over the horizon is

$$G_{\text{pref}}(\pi) = \sum_{k=t+1}^{t+H} \mathbb{E}_{q_{\pi}(y_k)} [-\log p_{\text{pref}}(y_k)]. \quad (12)$$

This term penalizes future observations that are unlikely under the preference distribution.

6.2 Information Gain or Epistemic Component

The information gain expected from future observations is

$$I_k(\pi) = \mathbb{E}_{q_{\pi}(y_k)} \left[D_{\text{KL}} \left(q_{\pi}(x_k | y_k) \| q_{\pi}(x_k) \right) \right]. \quad (13)$$

This quantity measures how much the future observation y_k is expected to reduce uncertainty about the hidden state x_k .

Because Active Inference minimizes expected free energy, information gain enters with a negative sign:

$$G_{\text{info}}(\pi) = - \sum_{k=t+1}^{t+H} I_k(\pi) \quad (14)$$

$$= - \sum_{k=t+1}^{t+H} \mathbb{E}_{q_{\pi}(y_k)} \left[D_{\text{KL}} \left(q_{\pi}(x_k | y_k) \| q_{\pi}(x_k) \right) \right]. \quad (15)$$

6.3 Combined Form

The trajectory-based expected free energy may therefore be written as

$$G(\pi) = \sum_{k=t+1}^{t+H} \mathbb{E}_{q_\pi(y_k)} [-\log p_{\text{pref}}(y_k)] - \sum_{k=t+1}^{t+H} \mathbb{E}_{q_\pi(y_k)} \left[D_{\text{KL}}(q_\pi(x_k | y_k) \| q_\pi(x_k)) \right] \quad (16)$$

subject to the predictive state and observation dynamics induced by U_π .

Equivalently, minimizing $G(\pi)$ is approximately equivalent to maximizing

$$\sum_{k=t+1}^{t+H} [\mathbb{E}_{q_\pi(y_k)} \log p_{\text{pref}}(y_k) + I_k(\pi)].$$

In words:

Choose future control sequences that produce preferred observations and informative observations.

7 Risk-Ambiguity Form

Another common decomposition is

$$G_k(\pi) = D_{\text{KL}}(q_\pi(y_k) \| p_{\text{pref}}(y_k)) + \mathbb{E}_{q_\pi(x_k)} [H(p(y_k | x_k))]. \quad (17)$$

The first term is often called risk. It measures the divergence between predicted observations and preferred observations. The second term is ambiguity. It penalizes states that produce intrinsically uncertain observations.

The full horizon version is

$$G(\pi) = \sum_{k=t+1}^{t+H} \{ D_{\text{KL}}(q_\pi(y_k) \| p_{\text{pref}}(y_k)) + \mathbb{E}_{q_\pi(x_k)} [H(p(y_k | x_k))] \}. \quad (18)$$

This form emphasizes risk and ambiguity; the previous form emphasizes expected preference satisfaction and information gain.

8 Prior and Posterior Over Policies

In Friston’s formulation, policies are treated as random variables. The prior probability of a policy is constructed from expected free energy:

$$p(\pi) = \sigma(-\gamma G(\pi)), \quad (19)$$

where γ is a precision or inverse-temperature parameter, and σ denotes the softmax function. For a finite policy set $\{\pi_1, \dots, \pi_N\}$,

$$p(\pi_i) = \frac{\exp[-\gamma G(\pi_i)]}{\sum_{j=1}^N \exp[-\gamma G(\pi_j)]}. \quad (20)$$

Policies with lower expected free energy have higher prior probability.

After observations have been obtained, each policy also has a variational free energy term $F(\pi)$ measuring how well that policy explains the observations. The posterior over policies is then

$$q(\pi_i) = \frac{\exp[-F(\pi_i) - \gamma G(\pi_i)]}{\sum_{j=1}^N \exp[-F(\pi_j) - \gamma G(\pi_j)]}. \quad (21)$$

Thus policy selection combines:

- explanation of observations already obtained, through $F(\pi)$;
- evaluation of future consequences, through $G(\pi)$.

9 Comparison of F and G

Quantity	Role	Time Direction
$F[q]$	Approximate inference over hidden states	Past and present
$G(\pi)$	Evaluation of future policies	Future

Variational free energy F

$$F[q] = D_{\text{KL}}(q(x)||p(x)) - \mathbb{E}_q[\log p(y | x)].$$

It asks:

What hidden state best explains the observed data?

Expected free energy $G(\pi)$

$$G(\pi) = \sum_{k=t+1}^{t+H} \mathbb{E}_{q_{\pi}(y_k)} [-\log p_{\text{pref}}(y_k)] - \sum_{k=t+1}^{t+H} I_k(\pi).$$

It asks:

Which future control sequence should be pursued?

10 Engineering Summary

Friston's compact notation hides a structure familiar from finite-horizon stochastic control. Written explicitly, the planning problem involves:

$$U_{\pi} = [u_t, u_{t+1}, \dots, u_{t+H-1}]^T, \quad (22)$$

$$X_{\pi} = [x_{t+1}, x_{t+2}, \dots, x_{t+H}]^T, \quad (23)$$

$$Y_{\pi} = [y_{t+1}, y_{t+2}, \dots, y_{t+H}]^T. \quad (24)$$

The expected free energy is a trajectory functional:

$$G(\pi) = G(U_{\pi}; X_{\pi}, Y_{\pi}).$$

The essential content is:

$$G(U_{\pi}) = \text{expected preference cost} - \text{expected information gain}$$

subject to the predictive dynamics induced by U_π .

This makes the analogy with model predictive control transparent, while preserving the distinctive Active Inference elements: preference distributions, information gain, and Bayesian inference over policies.